

Business Summary Report

Predictive Insights for Geldium's Collections Strategy

Prepared for: Head of Collections | Tata iQ GenAI Powered Data Analytics Forage virtual experience (simulated client engagement)

Executive Recommendation

Launch a 6-week pilot using transparent, segment-based rules (high debt-to-income ratio and unemployed customers) with personalized SMS/email outreach and hardship support, while continuing to improve the predictive model before relying on it for production decisions.

1. Summary of Predictive Insights

Our analysis of Geldium's 500-customer dataset identified which customer segments carry elevated delinquency risk, and evaluated whether a predictive model could reliably flag them in advance. The honest headline for this iteration: the tested model does not yet outperform simple segment-based rules, so the recommendation below is built directly on the segment patterns from the data rather than on model risk scores.

Key insight summary

Insight	Segment	Delinquency Rate	vs. Overall (16.0%)
Higher debt-to-income ratio tracks with higher delinquency	DTI $\geq$ 35%	18.5%	+2.5 pts
Unemployed customers delinquent well above retirees	Unemployed	19.4%	+3.4 pts
Combined high-DTI and unemployed customers are the highest-risk group found	DTI $\geq$ 35% AND Unemployed	30.4%	+14.4 pts
Mid-life customers (46–60) show elevated risk vs. neighboring age bands	Age 46–60	21.7%	+5.7 pts

Note on the model: a logistic regression was built and tested on this dataset (see Task 2) and returned an AUC of 0.44 — essentially no better than chance. This means individual risk scores from that model should not yet be used to prioritize outreach. The segment patterns above, however, are directly observed in the data (see Task 1, Section 4.4 for the combined Employment $\times$ DTI cross-tab underlying the 30.4% figure) and are usable today as a simpler, rules-based targeting approach while the model is improved.

2. Risk Segments and Recommended Actions

Translating the insights above into a simple, actionable tiering scheme for the collections team:

Segment	Risk Tier	Recommended Action
DTI $\geq$ 35% AND Unemployed	High	Payment plan offer + hardship support outreach
DTI $\geq$ 35% (all employment statuses)	Medium	Proactive SMS + email reminder

Segment	Risk Tier	Recommended Action
All other customers	Low	Routine monitoring, standard reminders only

3. Recommendation Framework

Restated insight

Customers who are both unemployed and carrying a debt-to-income ratio of 35% or higher show a 30.4% delinquency rate (see Task 1, Section 4.4) — nearly double the portfolio average — though this is a small group (23 customers in the current sample) and the pattern should be validated on a larger population before scaling significantly.

Proposed recommendation

Launch a 6-week pilot outreach campaign targeting the combined high-DTI / unemployed segment, plus the broader DTI $\geq$ 35% segment (142 customers) as a secondary tier, using SMS and email reminders paired with proactive payment-plan and hardship-support offers — rather than standard collections language.

SMART justification

- **Specific:** Targets a precisely defined segment (DTI $\geq$ 35%, with a priority sub-tier for unemployed customers) rather than blanket outreach.
- **Measurable:** Track delinquency rate within the pilot segment against a holdout group; target a reduction from the observed 18.5%–30.4% range toward the portfolio average of 16.0% or better within the pilot window.
- **Actionable:** Uses existing SMS/email infrastructure and payment-plan offers already available to the collections team; no new systems required for the pilot.
- **Relevant:** Directly addresses Geldium's stated goals of reducing delinquency and shifting from reactive to proactive collections.
- **Time-bound:** 6-week pilot with a defined review point before any wider rollout.

4. Expected KPIs

Success of the pilot should be measured against the following indicators, tracked weekly against a holdout group:

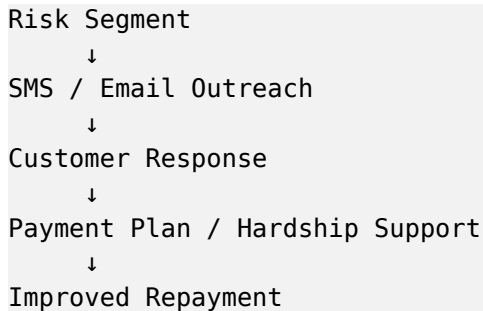
KPI	Target
Delinquency rate in pilot segment	Reduce from 18.5%–30.4% toward the 16.0% portfolio average
SMS / email response rate	Increase vs. current baseline outreach
Payment plan adoption rate	Increase among contacted high-risk customers
Customer engagement / satisfaction	Improve, measured via post-contact survey or complaint rate

5. Expected Business Value

- Reduced collection costs through earlier, better-targeted intervention
- Improved customer retention by pairing outreach with supportive options (payment plans, hardship support) rather than punitive action
- Better prioritization of the highest-risk customers instead of uniform outreach

- Improved operational efficiency for the collections team by focusing effort on the segments most likely to need it
- Fairer, more consistent customer treatment through transparent, auditable segment rules rather than an unvalidated model score

6. Pilot Workflow



7. Ethical and Responsible AI Considerations

Because the current model does not perform reliably (Section 1), this pilot is deliberately built on transparent, auditable segment rules rather than an opaque model score — which is itself a responsible AI choice: don't automate decisions on a model that hasn't demonstrated it works.

- **Bias:** Employment status and age are both being used as targeting criteria here, which requires care — these are sensitive in a lending context. The pilot should be monitored to confirm it results in supportive outreach (payment plans, hardship options) rather than any action that could disadvantage protected groups (e.g., credit line reductions).
- **Explainability:** Segment rules (DTI threshold, employment status) are fully transparent and easy to explain to both customers and regulators, which is preferable to a black-box score at this stage.
- **Human review:** Given the small size of the highest-risk sub-segment (23 customers), individual case review is feasible and recommended before any account-level action beyond routine reminders.
- **Responsible use:** All actions in this pilot are supportive (payment plans, hardship support, reminders) rather than punitive, consistent with fair-lending principles and Geldium's stated goal of improving customer outcomes alongside collections performance.

Looking ahead, future versions of this collections strategy can leverage Agentic AI to autonomously prioritize outreach and adapt messaging in real time, while preserving mandatory human oversight for hardship approvals, debt restructuring, and any high-impact or model-informed decision (see Task 4 for the full architecture).

8. Next Steps: Phased Roadmap

Phase 1 — Pilot

- Run the 6-week pilot on the defined segments using transparent rules-based targeting.

Phase 2 — Measurement

- Collect outcomes: delinquency rate, response rate, and payment-plan uptake against the KPIs in Section 4.

Phase 3 — Model Improvement

- Retrain the predictive model with non-linear approaches or engineered interaction terms, and validate against a larger or real production dataset (see Task 2).

Phase 4 — Responsible AI Validation

- Confirm fairness, explainability, and bias checks hold once the model is reintroduced, before it is allowed to inform any targeting decision.

Phase 5 — Scale

- Expand from pilot segments to the full portfolio once both the rules-based approach and the improved model have demonstrated reliable, fair performance.

9. Conclusion

The current predictive model is not sufficiently reliable for automated decision-making; however, transparent segment-based rules provide an immediately actionable strategy. A phased pilot, supported by Responsible AI principles and continuous model improvement, offers Geldium a practical path toward an AI-powered collections system.